

EMAIL_SETUP

Email (SMTP) Integration Setup Guide

This guide will walk you through setting up email notifications for HelixCode using SMTP.

Prerequisites

- SMTP server access (Gmail, Office 365, or custom server)
- Email account credentials or app password

Supported Email Providers

- **Gmail** (recommended for personal use)
- **Office 365 / Outlook**
- **Custom SMTP servers**
- **SendGrid, Mailgun** (transactional email services)

Setup for Gmail

Step 1: Enable 2-Factor Authentication

1. Go to <https://myaccount.google.com/security>
2. Enable “**2-Step Verification**”

Step 2: Generate App Password

1. Go to <https://myaccount.google.com/apppasswords>
2. Select “**Mail**” as the app
3. Select your device or enter custom name: HelixCode
4. Click “**Generate**”
5. **Copy the 16-character password** (e.g., abcd efgh ijkl mnop)

Step 3: Configure HelixCode

Add to your .env file:

```
HELIX_EMAIL_SMTP_SERVER=smtp.gmail.com  
HELIX_EMAIL_SMTP_PORT=587  
HELIX_EMAIL_USERNAME=your-email@gmail.com
```

```
HELIX_EMAIL_PASSWORD=abcdefghijklmnop # App password (no spaces)
HELIX_EMAIL_FROM=your-email@gmail.com
HELIX_EMAIL_RECIPIENTS=admin@example.com,team@example.com
```

Setup for Office 365 / Outlook

Step 1: Get Credentials

- Use your Office 365 email and password
- Ensure SMTP is enabled for your account (check with admin)

Step 2: Configure HelixCode

Add to your .env file:

```
HELIX_EMAIL_SMTP_SERVER=smtp.office365.com
HELIX_EMAIL_SMTP_PORT=587
HELIX_EMAIL_USERNAME=your-email@company.com
HELIX_EMAIL_PASSWORD=your-password
HELIX_EMAIL_FROM=your-email@company.com
HELIX_EMAIL_RECIPIENTS=admin@company.com
```

Setup for Custom SMTP Server

Step 1: Get SMTP Details

Contact your email administrator for: - SMTP server address - SMTP port (usually 587 for TLS, 465 for SSL, 25 for unencrypted) - Authentication credentials

Step 2: Configure HelixCode

Add to your .env file:

```
HELIX_EMAIL_SMTP_SERVER=smtp.yourcompany.com
HELIX_EMAIL_SMTP_PORT=587
HELIX_EMAIL_USERNAME=notifications@yourcompany.com
HELIX_EMAIL_PASSWORD=your-password
HELIX_EMAIL_FROM=HelixCode <notifications@yourcompany.com>
HELIX_EMAIL_RECIPIENTS=team@yourcompany.com
```

Configure config.yaml

Edit config/config.yaml:

```
notifications:
  enabled: true
  channels:
    email:
      enabled: true
      smtp:
        server: "${HELIX_EMAIL_SMTP_SERVER}"
        port: 587
        username: "${HELIX_EMAIL_USERNAME}"
        password: "${HELIX_EMAIL_PASSWORD}"
        from: "HelixCode <${HELIX_EMAIL_FROM}>"
        tls: true
      recipients:
        default: ["admin@example.com"]
        critical: ["admin@example.com", "oncall@example.com"]
      timeout: 30
```

Test Integration

Using HelixCode CLI (if available)

```
helix notify test --channel email --message "Test email
notification"
```

Using API

```
curl -X POST http://localhost:8080/api/v1/notifications/test \
-H "Content-Type: application/json" \
-H "Authorization: Bearer YOUR_TOKEN" \
-d '{
  "channel": "email",
  "title": "Test Email Notification",
  "message": "Testing email integration from HelixCode",
  "type": "info",
  "metadata": {
    "recipients": ["admin@example.com"]
  }
}'
```

Using Go Code

```
import "dev.helix.code/internal/notification"

// Create notification engine
```

```

engine := notification.NewNotificationEngine()

// Register email channel
emailChannel := notification.NewEmailChannel(
    "smtp.gmail.com",
    587,
    "your-email@gmail.com",
    "app-password",
    "HelixCode <your-email@gmail.com>",
)
engine.RegisterChannel(emailChannel)

// Send test notification
testNotif := &notification.Notification{
    Title:    "Test Email",
    Message:  "Testing email integration",
    Type:     notification.NotificationTypeInfo,
    Metadata: map[string]interface{}{
        "recipients": []string{"admin@example.com"},
    },
}
err := engine.SendDirect(context.Background(), testNotif,
    []string{"email"})

```

Recipient Management

Single Recipient

```

Metadata: map[string]interface{}{
    "recipient": "admin@example.com", // Note: singular
    "recipient"
}

```

Multiple Recipients

```

Metadata: map[string]interface{}{
    "recipients": []string{ // Note: plural "recipients"
        "admin@example.com",
        "team@example.com",
        "oncall@example.com",
    },
}

```

Using Default Recipients from Config

Configure in config.yaml:

```
email:
  recipients:
    default: ["admin@example.com", "team@example.com"]
    critical: ["admin@example.com", "oncall@example.com",
              "cto@example.com"]
```

Then use rules to select recipient lists:

```
notifications:
  rules:
    - name: "Critical errors to oncall"
      condition: "type==error"
      channels: ["email"]
      priority: urgent
      template: "critical_alert"
```

Common SMTP Ports

Port	Protocol	Description	Recommended
25	SMTP (unencrypted)	Legacy, often blocked by ISPs	No
465	SMTPS (SSL)	Deprecated but still works	Use if required
587	SMTP + STARTTLS	Modern, secure	Recommended

Troubleshooting

Error: “Authentication failed”

- **Gmail:** Ensure you’re using an app password, not your account password
- **Office 365:** Verify your credentials and that SMTP is enabled
- **Custom:** Check username and password with your admin

Error: “Connection refused”

- **Cause:** Incorrect SMTP server or port
- **Solution:** Verify SMTP server address and port number
- **Gmail:** Use smtp.gmail.com:587
- **Office 365:** Use smtp.office365.com:587

Error: “TLS handshake failed”

- **Cause:** TLS/SSL configuration issue
- **Solution:**
 - Try port 465 with `tls: false` (uses SSL instead)
 - Verify server supports STARTTLS on port 587

Error: “Recipient not specified”

- **Cause:** Missing recipients in notification metadata
- **Solution:** Always include recipients in metadata:

```
Metadata: map[string]interface{}{  
    "recipients": []string{"admin@example.com"},  
}
```

Emails going to spam

- **Cause:** Missing SPF, DKIM, DMARC records
- **Solution:**
 - For Gmail/Office365: Emails should not go to spam
 - For custom domains: Configure SPF, DKIM, DMARC DNS records
 - Add “HelixCode” to subject to help filtering

Rate limiting

- **Gmail:** 500 emails/day (free), 2000/day (Workspace)
- **Office 365:** Varies by plan (typically 10,000/day)
- **Solution:** Use dedicated transactional email service for high volume

Advanced Configuration

HTML Email Templates (Future Feature)

Future versions will support HTML templates:

```
email:  
  templates:  
    enabled: true  
    html: true  
    path: "/etc/helixcode/email-templates"
```

Attachments (Future Feature)

Future versions will support file attachments:

```
Metadata: map[string]interface{}{
    "recipients": []string{"admin@example.com"},
    "attachments": []string{
        "/tmp/logs/error.log",
        "/tmp/reports/summary.pdf",
    },
}
```

Email Priority

Set email priority/importance:

```
email:
    priority_header: true # Adds X-Priority header
```

Security Best Practices

- 1. Never commit passwords to version control**
 - Use environment variables
 - Add .env to .gitignore
- 2. Use app passwords, not account passwords**
 - Gmail: Use app passwords
 - Office 365: Use app passwords where supported
- 3. Enable TLS/SSL**
 - Always use encrypted connections
 - Prefer port 587 with STARTTLS
- 4. Rotate credentials regularly**
 - Change passwords periodically
 - Regenerate app passwords
- 5. Use dedicated email accounts**
 - Don't use personal email for system notifications
 - Create dedicated accounts like notifications@company.com
- 6. Implement SPF/DKIM/DMARC (for custom domains)**
 - Prevents email spoofing
 - Improves deliverability
 - Reduces spam classification

Using Transactional Email Services

For high-volume or production use, consider dedicated services:

SendGrid

```
HELIX_EMAIL_SMTP_SERVER=smtp.sendgrid.net  
HELIX_EMAIL_SMTP_PORT=587  
HELIX_EMAIL_USERNAME=apikey  
HELIX_EMAIL_PASSWORD=your-sendgrid-api-key
```

Mailgun

```
HELIX_EMAIL_SMTP_SERVER=smtp.mailgun.org  
HELIX_EMAIL_SMTP_PORT=587  
HELIX_EMAIL_USERNAME=postmaster@your-domain.mailgun.org  
HELIX_EMAIL_PASSWORD=your-mailgun-password
```

Amazon SES

```
HELIX_EMAIL_SMTP_SERVER=email-smtp.us-east-1.amazonaws.com  
HELIX_EMAIL_SMTP_PORT=587  
HELIX_EMAIL_USERNAME=your-ses-smtp-username  
HELIX_EMAIL_PASSWORD=your-ses-smtp-password
```

Resources

- [Gmail SMTP Settings](#)
- [Office 365 SMTP Settings](#)
- [SMTP RFC Documentation](#)
- [HelixCode Notification API Documentation](#)

Support

If you encounter issues: 1. Check the [Troubleshooting](#) section 2. Verify SMTP settings with your email provider 3. Check HelixCode logs for detailed error messages 4. Test SMTP connection with tools like telnet or swaks 5. Open an issue on GitHub